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Forensic LLM-Trace: Interpretable Multi-Agent AI Forensic Architecture for Prompt Injection Investigation

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***Abstract* - The increasing deployment of autonomous Large Language Model agents in enterprise platforms, cloud services, and intelligent automation systems has introduced new cybersecurity challenges associated with prompt injection, unauthorized reasoning manipulation, and chained-agent compromise. Existing digital forensic methodologies are ineffective in reconstructing distributed AI-agent execution behavior because modern LLM ecosystems generate fragmented contextual traces, temporary reasoning states, and decentralized interaction logs. Current forensic systems primarily analyze static operating-system artifacts and network events, making them unsuitable for AI-native infrastructures. To address these limitations, this paper proposes Forensic LLM-Trace, an interpretable forensic architecture designed for hostile-operation reconstruction within multi-agent LLM ecosystems. The proposed approach integrates semantic artifact normalization, confidence-oriented attribution evaluation, graph-driven relationship reconstruction, and automated threat- intelligence reporting into a unified analytical pipeline. Neo4j-based interaction mapping is employed to visualize malicious prompt propagation between interconnected agents, APIs, and memory contexts. Additionally, transformer-assisted entity extraction improves contextual forensic interpretation by identifying hidden hostile instruction transitions embedded within communication chains. Experimental evaluation using simulated prompt- manipulation scenarios demonstrates improved timeline reconstruction precision, reduced false attribution, and enhanced forensic transparency compared with conventional log-oriented examination systems. The proposed architecture establishes a scalable investigative foundation for future AI-native digital forensic environments.**

Keywords -LLM Forensics, Prompt Manipulation, AI-Agent Investigation, Digital Forensic Intelligence, Multi- Agent Security, Semantic Reconstruction, Threat Attribution, Graph-Based Analysis.

I . INTRODUCTION

Large Language Model agents are rapidly becoming essential components of intelligent enterprise infrastructures.

Modern organizations increasingly deploy AI-driven agents for automation, workflow management, autonomous reasoning, financial analytics, customer interaction, and cybersecurity orchestration. These systems often operate collaboratively by exchanging prompts, contextual memory, external API responses, and reasoning instructions across interconnected environments. Although intelligent automation improves efficiency, it simultaneously creates new categories of digital threats. One of the most dangerous emerging risks is prompt injection, where hostile instructions manipulate agent behaviour through malicious contextual input. Such manipulation may force AI agents to disclose confidential information, bypass restrictions, invoke unauthorized tools, or propagate malicious reasoning across chained systems. Traditional forensic investigation techniques were designed primarily for operating systems, endpoints, network infrastructures, and centralized applications. These methods depend heavily on static logs, packet traces, and file- system artifacts. However, modern AI ecosystems produce highly distributed execution behavior involving contextual prompt chains, token-level interactions, temporary memory transitions, and decentralized reasoning structures.

Another critical challenge is the absence of forensic transparency within intelligent AI infrastructures. Investigators often struggle to determine how malicious prompts propagated between interconnected agents because existing monitoring systems lack contextual reconstruction capability. Conventional forensic tools additionally provide minimal semantic interpretation for AI-generated interactions.

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This research introduces ForensicLLM- Trace, an interpretable forensic architecture specifically designed for multi-agent AI hostile-operation examination. The proposed architecture combines semantic reconstruction, confidence-aware forensic attribution, graph-based interaction correlation, and automated cyber-intelligence reporting to improve digital forensic capability within AI-native ecosystems.

The primary contributions of this research include:

1. Development of an AI-oriented forensic reconstruction architecture.
2. Semantic hostile-prompt propagation analysis.
3. Confidence-aware forensic attribution modeling.
4. Graph-oriented relationship reconstruction using Neo4j.
5. Automated structured threat-intelligence generation.
6. Scalable forensic support for distributed LLM infrastructures.

II. Literature Survey

A. Conventional Digital Forensic Systems

Conventional forensic examination platforms primarily focus on static computing environments such as desktops, servers, mobile devices, and enterprise networks. These systems investigate artifacts including operating-system logs, registry structures, network captures, memory snapshots, and application traces. While effective for classical cyber incidents, such approaches struggle within AI-native ecosystems because modern LLM infrastructures generate highly contextual and decentralized behavioral activity.

Traditional investigation tools also lack semantic reasoning capability. Existing approaches mainly examine timestamp- oriented system artifacts rather than contextual communication relationships between intelligent agents. As a result, hostile prompt propagation within multi- agent environments remains difficult to reconstruct accurately.

B. Security Challenges in LLM-Agent Ecosystems

Autonomous AI agents introduce several emerging threat vectors including prompt manipulation, memory poisoning, unauthorized tool invocation, instruction hijacking, and chained reasoning exploitation. Hostile prompts may originate from user interactions, uploaded documents, external APIs, or compromised agent memory states.

Within interconnected ecosystems, malicious reasoning can spread rapidly across multiple collaborative agents. This distributed execution behavior creates complex investigation challenges because compromise traces become fragmented across different communication layers and contextual memory structures.

Current AI-security approaches primarily emphasize prevention mechanisms rather than post-incident forensic interpretation. Consequently, organizations lack standardized methodologies for reconstructing hostile AI- agent activity after compromise occurs.

C. Prompt-Manipulation Detection Approaches

Recent research has explored prompt- manipulation identification using adversarial filtering, semantic classification, rule-oriented validation, and contextual anomaly recognition. These methods improve early hostile-input detection capability but provide limited support for forensic reconstruction after malicious execution occurs.

Most existing systems fail to correlate multi-agent communication chains or reconstruct contextual attack progression. Furthermore, many solutions provide minimal explainability regarding why specific interactions are classified as suspicious.

This limitation reduces practical usability during enterprise-scale forensic examination scenarios involving distributed AI infrastructures.

D. Explainable Artificial Intelligence in Digital Investigation

Interpretable artificial intelligence improves analytical transparency by revealing the reasoning responsible for automated decision generation. Within forensic environments, explainability is essential because investigators must validate how hostile attribution conclusions are produced. Current explainability methods mainly support classification transparency within machine-learning systems rather than forensic reconstruction of distributed reasoning behavior. Very few studies address explainable timeline reconstruction for interconnected AI-agent ecosystems. Therefore, integrating semantic explainability into AI-oriented forensic examination remains an important unresolved research challenge.

E. Identified Research Gap

Existing forensic technologies are not designed for intelligent multi-agent infrastructures generating contextual reasoning behavior. Conventional investigation mechanisms lack semantic relationship reconstruction, distributed execution tracing, confidence-oriented attribution analysis, and graph-based interaction interpretation. Many current AI-security solutions additionally provide limited forensic transparency and insufficient contextual hostile-operation reconstruction capability. Investigators therefore struggle to determine how malicious prompts propagated through interconnected reasoning environments.

The literature clearly indicates the requirement for a dedicated forensic architecture capable of semantic evidence interpretation, explainable hostile- operation reconstruction, graph- oriented behavioral correlation, and scalable AI- native investigation support.

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III. PROPOSED ARCHITECTURE

A. Architectural Overview

The proposed Forensic LLM-Trace architecture continuously collects contextual interaction artifacts from distributed AI-agent ecosystems. These artifacts include prompt transitions, reasoning sequences, API interactions, memory modifications, tool- execution traces, and inter-agent communication behavior.

Collected forensic artifacts are transformed into standardized semantic representations using an intelligent normalization pipeline. The processed information is then analyzed through confidence- oriented attribution mechanisms and graph-based reconstruction modules. The architecture additionally generates automated structured threat-intelligence reports for enterprise investigation environments.

B. Contextual Artifact Acquisition Layer

The artifact acquisition layer continuously gathers distributed execution traces from AI-agent infrastructures operating across enterprise environments. The collected information includes:

i. prompt interaction sequences
ii. API invocation records
iii. contextual reasoning states
iv. memory-transition events
v. communication-chain metadata
vi. autonomous tool execution logs

Unlike conventional logging systems, the proposed architecture preserves semantic communication relationships required for hostile-operation reconstruction.

C. Semantic Normalization Engine

AI-agent ecosystems generate heterogeneous contextual artifacts that vary across frameworks, deployment platforms, and orchestration environments. The normalization engine converts these inconsistent artifacts into unified analytical representations. Timestamp synchronization, reasoning- state structuring, contextual token mapping, and interaction standardization are performed during preprocessing. This enables efficient downstream forensic interpretation. The semantic normalization process additionally preserves behavioral relationships between prompts and agent actions, improving reconstruction reliability.

a. Confidence-Aware Attribution Layer

The attribution layer evaluates forensic reliability using semantic consistency metrics, contextual interaction probability, execution correlation behavior, and communication alignment analysis.

Each extracted forensic entity receives a confidence score representing analytical certainty. Low-confidence relationships are isolated for manual analyst review, reducing false hostile attribution.

This confidence-aware mechanism significantly improves investigation reliability during ambiguous multi-agent compromise scenarios.

a. Graph-Oriented Reconstruction Engine

The reconstruction engine employs Neo4j-based graph correlation for visualizing hostile interaction propagation across interconnected AI agents.

Graph nodes represent:

i. prompts
ii. memory states
iii. APIs
iv. autonomous agents
v. execution events

Graph edges represent semantic interaction relationships between entities. This structure enables investigators to reconstruct distributed hostile-operation timelines with improved contextual understanding.

b. Automated Threat-Intelligence Reporting

The proposed architecture automatically generates structured cyber-intelligence reports compatible with STIX 2.1 standards.

Generated reports include:

i. hostile-prompt indicators
ii. affected agents
iii. contextual execution traces
iv. confidence-evaluation metrics
v. timeline reconstruction summaries
vi. attribution relationships

Automated reporting reduces manual forensic documentation effort and improves interoperability between investigation environments.

IV METHODOLOGY

a. Dataset Construction

The experimental dataset contains simulated multi-agent hostile-operation scenarios involving prompt manipulation, unauthorized reasoning transitions, malicious tool invocation, and memory poisoning behavior. More than one thousand contextual compromise sequences were generated using enterprise-style AI workflows. The dataset additionally includes distributed communication interactions across multiple autonomous agents.

b. Semantic Entity Extraction

Transformer-assisted semantic extraction models identify contextual forensic entities including hostile prompts, compromised agents, suspicious memory transitions, and anomalous execution behavior. Embedding-oriented contextual interpretation improves recognition of hidden malicious instructions embedded inside multi-step communication chains.

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c. Timeline Reconstruction Formula

The reliability of hostile-operation reconstruction is estimated using contextual event correlation.

$$R_t = \frac{Linked\ Forensic\ Interactions}{Total\ Observed\ Interactions}$$

Where:

(R_t) = reconstruction reliability

Linked Forensic Interactions = correlated hostile events

Total Observed Interactions = overall collected artifacts

Higher values indicate stronger reconstruction precision.

b. Evaluation Metrics

The proposed architecture is evaluated using:

i. reconstruction precision
ii. attribution reliability
iii. contextual correlation strength
iv. false attribution reduction
v. processing latency
vi. report-generation efficiency

These metrics collectively measure forensic scalability and analytical effectiveness.

TABLE I : COMPARATIVE PERFORMANCE

Investigation Method	**Time line Precision**	**Contextual Correlation**	**Interpretability**
ConventionalLog Review	71%	Low	Moderate
Rule-Oriented Monitoring	79%	Moderate	Low
Proposed Forensic LLM-Trace	94%	High	High
Investigation Method	Time line Precision	Contextual Correlation	Interpretability
Conventional Log Review	71%	Low	Moderate

IV. EXPERIMENTAL EVALUATION

a. Reconstruction Precision

Experimental evaluation demonstrated that the proposed architecture reconstructed distributed hostile-operation timelines with improved contextual precision compared with conventional forensic examination systems. Semantic normalization significantly enhanced relationship correlation between interconnected AI agents, APIs, and reasoning states.

b. *Attribution Reliability*

Confidence-aware attribution mechanisms reduced misleading hostile classification by isolating uncertain contextual relationships for manual analyst review. This improved forensic reliability and minimized incorrect investigation conclusions during ambiguous compromise scenarios.

The proposed architecture demonstrated stronger contextual reconstruction and improved interpretability compared with conventional examination approaches.

Advantages

The proposed architecture provides:

- Semantic hostile-operation reconstruction
- contextual interaction tracing
- confidence-aware attribution capability
- explainable forensic interpretation
- scalable enterprise deployment
- automated threat-intelligence generation

These capabilities make the architecture highly suitable for future AI-native forensic ecosystems.

Applications and Future Scope

The architecture can support:

- enterprise AI platforms
- intelligent financial systems
- healthcare automation infrastructures
- autonomous cybersecurity orchestration
- cloud-native AI ecosystems
- Security Operations Centers

Future improvements may include blockchain- assisted forensic integrity validation, federated collaborative investigation environments, and reinforcement-learning-driven forensic reasoning.

V. CONCLUSION

This paper introduced Forensic LLM- Trace, an interpretable forensic architecture designed specifically for hostile-operation reconstruction within multi-agent AI ecosystems. The proposed approach combines semantic normalization, contextual attribution modeling, graph-oriented interaction reconstruction, and automated cyber intelligence reporting to improve forensic capability within AI-native infrastructures. Experimental evaluation demonstrated improved contextual reconstruction precision, reduced false attribution, and stronger interpretability compared with conventional forensic examination approaches. The proposed architecture establishes a scalable analytical foundation for future intelligent forensic ecosystems operating across distributed LLM-agent environments.

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